Observational Uniqueness for Zero-Parameter Frameworks

Quotient Collapse Under the Admissibility Posture,
and the Open Programme for Full Inevitability

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Abstract

We prove that any physics framework satisfying five structural constraints—inhabitedness, zero adjustable parameters, self-similarity, an admissible cost functional obeying a coherent composition law, and observables extracted via cost minimization—is forced into a unique observational state. Formally: the quotient of the state space by observational indistinguishability collapses to a singleton. The preferred scale is forced to the golden ratio $\varphi = (1 + \sqrt{5})/2$, and the cost functional is uniquely $J_{\text{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Three of the five assumptions (self-similarity, the composition law, and cost-minimum semantics) are *adopted* axioms whose derivation from the zero-parameter posture alone is an open programme. We name six structural bridges that would close this programme, characterize why each is hard, and identify the derivation of the composition law from ledger structure as the deepest open problem. The composition law is not a bespoke equation: we show it is equivalent to the d'Alembert functional equation, classified by Aczél (1966).

All proved results are formally verified in Lean 4 with zero `sorry`. Core results are conditional on one external axiom (the Aczél classification theorem for continuous d'Alembert solutions).

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1 Introduction

1.1 The question

The Standard Model of particle physics has its symmetry group fixed by gauge consistency, but its 19 free parameters—coupling constants, masses, mixing angles—must be measured, not derived. What would happen if one demanded *zero* adjustable dimensionless constants?

This paper answers a precise version of that question. We define a minimal interface for a "physics framework" (a state space, evolution, and measurement map), formalize "zero parameters" as algorithmic describability of the state space, and prove:

> *Under five explicit structural assumptions, every admissible framework has exactly one observational state on the quotient.*

The five assumptions divide into two tiers (Section 3): two that are forced by the zero-parameter posture (Tier 1), and three that are adopted as structural axioms whose derivation from Tier 1 alone is an open programme (Tier 2).

1.2 What this paper proves and what it does not

Level	Proved	Open
Tier 1 (A1)+(A2)	Discreteness, comparison structure, countable skeleton	—
Bridges B1–B4	Fibonacci minimality among self-similar schemas	(HS)⇒B1; RCL derivation (B2, deepest); ratio interface B3; ground-state B4
Tier 2 (A1)–(A5)	$\sigma = \varphi$, $J = J_{\text{cost}}$, μ_F uniform, $\mathrm{SQ}(F) \simeq \mathbf{1}$	—
Reduction	Both F and canonical framework have trivial quotient	B5: prediction map; B6: value equality

The core contribution is the "Tier 2" row. Full inevitability requires closing bridges B1–B6 (Section 9).

1.3 Relation to existing work

Wigner (1939) classified particles by Poincaré representations. Coleman–Mandula (1967) showed the S-matrix symmetry group is uniquely determined. Both share the logic: *structural constraints collapse the space of admissible theories.* Our result is of this type, applied to the zero-parameter posture rather than to relativistic symmetry.

The cost-uniqueness theorem (T5) was proved independently by Washburn–Zlatanović [2] and Washburn–Rahnamai Barghi [3]; the d'Alembert reduction was formalized in [4]. The classical reference for the d'Alembert functional equation is Aczél [1].

1.4 Roadmap

Section 2 defines the framework interface and the observational quotient. Section 3 states the five assumptions and their tier classification. Section 4 provides the necessity stack justifying why the

assumptions are not ad hoc. Section 5 exposes the d'Alembert structure of the composition law—the strongest argument that (A4) is forced. Section 6 proves cost uniqueness ($J = J_{\text{cost}}$). Section 8 proves the main theorem (quotient collapse). Section 9 names the six open bridges and characterizes the open derivation programme. Section 10 lists concrete falsifiers.

2 Frameworks and the Observational Quotient

Definition 2.1 (Physics Framework)**.** A *physics framework* is a tuple $F = (S_F, T_F, O_F, \mu_F)$ where S_F is a nonempty set (states), $T_F : S_F \to S_F$ is an evolution operator, O_F is a set (observables), and $\mu_F : S_F \to O_F$ is a measurement map.

Definition 2.2 (Algorithmic Specification)**.** A set S admits an *algorithmic specification* if there exist a finite binary string d, an enumeration $g : \mathbb{N} \to \{0,1\}^* \cup \{\bot\}$, and a decoder $\psi : \{0,1\}^* \to S \cup \{\bot\}$ such that for every $s \in S$ there exist $n \in \mathbb{N}$ and $c \in \{0,1\}^*$ with $g(n) = c$ and $\psi(c) = s$.

Definition 2.3 (Zero-Parameter Framework)**.** F has *zero parameters* (written $F \in \mathcal{Z}$) if S_F admits an algorithmic specification.

Definition 2.4 (Observational Equivalence)**.** For a framework F, define $s_1 \sim_F s_2 \iff \mu_F(s_1) = \mu_F(s_2)$.

Proposition 2.5. *$\sim_F$ is an equivalence relation.*

Proof. Reflexivity, symmetry, and transitivity of equality. □

Definition 2.6 (Quotient State Space)**.** $\mathrm{SQ}(F) := S_F/{\sim_F}$, the set of observational equivalence classes. We say the quotient *collapses* when $\mathrm{SQ}(F)$ is a subsingleton (at most one element).

Lemma 2.7 (Uniform Observables Imply Quotient Collapse)**.** *If $\mu_F(s_1) = \mu_F(s_2)$ for all $s_1, s_2 \in S_F$, then $\mathrm{SQ}(F)$ is a subsingleton.*

Proof. Any two classes $[a], [b]$ satisfy $\mu_F(a) = \mu_F(b)$, hence $a \sim_F b$, hence $[a] = [b]$. □

3 The Five Assumptions

Definition 3.1 (Model-Independent Assumptions)**.** The *admissibility bundle* for a framework F consists of:

(A1) F is *inhabited*: $S_F \neq \emptyset$.

(A2) F has *zero parameters*: $F \in \mathcal{Z}$.

(A3) F admits a *self-similarity structure* (Theorem 4.4).

(A4) F has an *admissible cost functional* J satisfying the six conditions of Theorem 5.1.

(A5) F's observables are *extracted via cost* with all states at the cost minimum (Theorems 7.1 and 7.2).

Assumption	Status	Tier
(A1) Inhabited	Forced (any non-empty framework)	1
(A2) Zero parameters	Definitional (class $\mathcal{Z}$)	1
(A3) Self-similarity	Adopted; derivation from (A2) open	2
(A4) Cost + composition law	Adopted; derivation from ledger open	2
(A5) Observables from cost	Adopted; ratio interface + ground-state open	2

Tier 1 assumptions are consequences of the zero-parameter posture. Tier 2 assumptions import structural data whose derivation from (A1)–(A2) alone is the *open derivation programme* (Section 9). All theorems in this paper are conditional on the full bundle (A1)–(A5).

4 The Necessity Stack

Each Tier 2 assumption is justified—though not yet derived—by a chain of structural necessities:

$$\text{Observables} \Rightarrow \text{Comparison} \Rightarrow \text{Discrete} \Rightarrow \text{Conservation} \Rightarrow \varphi.$$

We summarize each arrow.

4.1 Observables require comparison

Proposition 4.1. *Any framework that derives non-trivial observables in a zero-parameter setting without external oracles must internally realize a reflexive, symmetric comparison mechanism that distinguishes states with different observable values.*

Argument. The measurement map μ_F is a function $S_F \to O_F$. If μ_F is non-constant, there exist s_1, s_2 with $\mu_F(s_1) \neq \mu_F(s_2)$. In a closed system with no external oracle, the detection of this inequality must be realized by an internal function $\delta : S_F \times S_F \to \{0, 1\}$ satisfying $\mu_F(s_1) \neq \mu_F(s_2) \Rightarrow \delta(s_1, s_2) = 1$. Reflexivity ($\delta(s, s) = 1$) and symmetry ($\delta(s_1, s_2) = \delta(s_2, s_1)$) follow from equality being an equivalence relation. □

4.2 Zero parameters force discreteness

Proposition 4.2. *Any $F \in \mathcal{Z}$ admits a surjection from a countable set onto S_F.*

Proof. The algorithmic specification provides an enumeration $g : \mathbb{N} \to \{0, 1\}^*$ and a decoder ψ covering every $s \in S_F$. The composite $\psi \circ g$ is a surjection from a countable index set. □

4.3 Discrete conservation forces balanced accounting

Proposition 4.3. *Given a locally finite discrete event system with a conserved flow, the conservation law is equivalent to a balanced double-entry condition on events.*

Sketch. Conservation at each event e requires $\sum_{v \to e} f(v) = \sum_{e \to w} f(w)$. This is precisely the balance condition of a double-entry accounting system. The discrete, locally finite structure ensures the sums are finite and well-defined. □

4.4 Self-similarity forces φ

Definition 4.4 (Self-Similarity Structure)**.** A framework admits a *self-similarity structure* if there exist a preferred scale $\sigma > 1$ and three successive levels $\ell_0, \ell_1, \ell_2 > 0$ with $\ell_1 = \sigma\ell_0,\ \ell_2 = \sigma\ell_1,$ and $\ell_2 = \ell_1 + \ell_0$.

Theorem 4.5 (Self-Similarity Forces φ)**.** *If F admits a self-similarity structure with scale σ, then*

$$\sigma^2 = \sigma + 1, \qquad \textit{hence} \qquad \sigma = \varphi := \frac{1+\sqrt{5}}{2}.$$

Proof. From the definitions: $\ell_2 = \sigma^2\ell_0$ and $\ell_2 = (\sigma+1)\ell_0$. Since $\ell_0 > 0$, cancel to get $\sigma^2 = \sigma + 1$. The polynomial $x^2 - x - 1$ has roots $(1 \pm \sqrt{5})/2$. Since $\sigma > 1$, we have $\sigma = (1+\sqrt{5})/2 = \varphi$. □

5 The d'Alembert Structure of the Composition Law

This section exposes the composition law (A4) as a classical functional equation, not a bespoke recognition-specific identity. This is the strongest argument that (A4) is not ad hoc.

5.1 The Recognition Composition Law

Definition 5.1 (Admissible Cost Functional)**.** A cost functional $J : \mathbb{R}_{>0} \to \mathbb{R}$ is *admissible* if:

1. *Reciprocal symmetry:* $J(x) = J(x^{-1})$ for all $x > 0$.
2. *Unit normalization:* $J(1) = 0$.
3. *Strict convexity:* J is strictly convex on $\mathbb{R}_{>0}$.
4. *Calibration:* $\frac{d^2}{dt^2}(J \circ \exp)\big|_{t=0} = 1$.
5. *Continuity:* J is continuous on $\mathbb{R}_{>0}$.
6. *Composition law (RCL):* For all $x, y > 0$,

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{1}$$

5.2 Reduction to the d'Alembert equation

Define the reparametrizations

$$G(t) := J(e^t), \qquad H(t) := G(t) + 1 = J(e^t) + 1. \tag{2}$$

Proposition 5.2 (RCL $\Leftrightarrow$ d'Alembert)**.** *J satisfies (1) if and only if H satisfies the d'Alembert functional equation:*

$$H(s+t) + H(s-t) = 2\,H(s)\,H(t), \qquad s, t \in \mathbb{R}. \tag{3}$$

Moreover, the admissibility conditions translate as follows:

- $J(1) = 0 \;\Leftrightarrow\; H(0) = 1$.
- $J(x) = J(x^{-1}) \;\Leftrightarrow\; H(-t) = H(t)$ *(evenness).*
- *Continuity of J on $\mathbb{R}_{>0}$ $\Leftrightarrow$ continuity of H on $\mathbb{R}$.*

Proof. Substitute $x = e^s$, $y = e^t$ into (1) and expand $K = J + 1$:

$$\begin{aligned}
J(e^{s+t}) + J(e^{s-t}) &= 2J(e^s)J(e^t) + 2J(e^s) + 2J(e^t) \\
G(s+t) + G(s-t) &= 2G(s)G(t) + 2G(s) + 2G(t) \\
(H(s+t) - 1) + (H(s-t) - 1) &= 2(H(s) - 1)(H(t) - 1) + 2(H(s) - 1) + 2(H(t) - 1) \\
H(s+t) + H(s-t) - 2 &= 2H(s)H(t) - 2 \\
H(s+t) + H(s-t) &= 2H(s)H(t).
\end{aligned}$$

□

5.3 The Aczél classification

The d'Alembert equation (3) is classical. Jean le Rond d'Alembert studied it in the eighteenth century in connection with vibrating strings. The complete classification of continuous solutions was established by Aczél [1]:

Theorem 5.3 (Aczél Classification [1, Ch. 3])**.** *The continuous solutions of $H(s+t) + H(s-t) = 2H(s)H(t)$ with $H(0) = 1$ are exactly:*

(i) $H(t) = 1$ *for all* t *(constant),*

(ii) $H(t) = \cosh(\lambda t)$ *for some* $\lambda \neq 0$*,*

(iii) $H(t) = \cos(\lambda t)$ *for some* $\lambda \neq 0$*.*

Remark 5.4 (Why the RCL is not ad hoc). The composition law (1) is not a bespoke equation engineered to produce a desired answer. It is the d'Alembert functional equation in multiplicative coordinates—the unique symmetric composition law on $(\mathbb{R}_{>0}, \times)$ compatible with non-degeneracy. The analogy with the parallelogram law is precise: Jordan–von Neumann (1935) proved that the parallelogram identity uniquely characterizes inner-product spaces among normed spaces; the d'Alembert identity uniquely characterizes cosh among symmetric composition functions on abelian groups.

5.4 Why additive composition fails

One might ask: why not $J(xy) = J(x) + J(y)$ (logarithmic cost)? Logarithms satisfy $J(x) = J(x^{-1})$ only if $J \equiv 0$, violating strict convexity. The d'Alembert form arises specifically from combining reciprocal symmetry with the requirement that the composition is quadratic (not additive) in J. Establishing *why* the composition must be quadratic rather than additive is Bridge B2 (Section 9.2).

6 Cost Uniqueness

Theorem 6.1 (T5: Cost Uniqueness)**.** *Let $J : \mathbb{R}_{>0} \to \mathbb{R}$ be an admissible cost functional (Theorem 5.1). Then for all $x > 0$:*

$$J(x) = J_{\text{cost}}(x) := \tfrac{1}{2}(x + x^{-1}) - 1. \tag{4}$$

Proof. By Theorem 5.2, $H(t) = J(e^t) + 1$ is a continuous, even solution of the d'Alembert equation with $H(0) = 1$.

Step 1: Apply the Aczél classification (Theorem 5.3). The three cases are: (i) $H = 1$, (ii) $H = \cosh(\lambda t)$, (iii) $H = \cos(\lambda t)$.

Step 2: Eliminate cases (i) and (iii). Case (i) gives $J \equiv 0$, contradicting strict convexity (which requires a unique minimum, not a flat zero). Case (iii) gives $J(e^t) = \cos(\lambda t) - 1 \leq 0$ with a maximum at $t = 0$; strict convexity requires a *minimum* at $t = 0$, ruling this out.

Step 3: Calibrate case (ii). $H(t) = \cosh(\lambda t)$ gives $\frac{d^2}{dt^2} H\big|_{t=0} = \lambda^2$. The calibration condition forces $\lambda^2 = 1$, hence $\lambda = 1$ (taking $\lambda > 0$).

Step 4: Conclude. $J(e^t) = \cosh(t) - 1 = \frac{1}{2}(e^t + e^{-t}) - 1$. Substituting $x = e^t$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1 = J_{\text{cost}}(x)$. □

Corollary 6.2. *$J_{\text{cost}}(x) = 0$ if and only if $x = 1$. J_{cost} is strictly convex on $\mathbb{R}_{>0}$ with unique minimum at $x = 1$.*

Proof. $J_{\text{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1 \geq 0$ by AM–GM, with equality iff $x = x^{-1}$, i.e., $x = 1$ (since $x > 0$). □

7 Observables from Cost and Derived Uniformity

Definition 7.1 (Measure Determined by Cost)**.** F has *measure determined by cost* J if there exists $r : S_F \to \mathbb{R}_{>0}$ (the ratio extraction) such that

$$r(s_1) = r(s_2) \implies \mu_F(s_1) = \mu_F(s_2).$$

Definition 7.2 (Observables from Cost)**.** F has *observables from cost* if it has measure determined by cost and every state lies at the cost minimum: $J(r(s)) = 0$ for all $s \in S_F$.

Theorem 7.3 (Derived Uniformity)**.** *Let F be a framework with an admissible cost functional J, measure determined by cost via $r : S_F \to \mathbb{R}_{>0}$, and $J(r(s)) = 0$ for all s. Then:*

1. *$J = J_{\text{cost}}$ (by T5).*
2. *$r(s) = 1$ for all s (by Theorem 6.2).*
3. *μ_F is uniform: $\mu_F(s_1) = \mu_F(s_2)$ for all s_1, s_2.*

Proof. (1) Theorem 6.1. (2) $J_{\text{cost}}(r(s)) = 0$ and $J_{\text{cost}}(x) = 0 \Leftrightarrow x = 1$ (Theorem 6.2). (3) $r(s_1) = 1 = r(s_2)$, so measure-determined-by-cost gives $\mu_F(s_1) = \mu_F(s_2)$. □

8 Main Theorem

Theorem 8.1 (Observational Uniqueness on the Quotient)**.** *Let F be a physics framework satisfying* (A1)–(A5) *of Theorem 3.1. Then:*

1. *The preferred scale is $\varphi = (1 + \sqrt{5})/2$.*
2. *The cost functional is $J = J_{\text{cost}}$.*
3. *The measurement map is uniform.*
4. *The quotient state space is a subsingleton: $\mathrm{SQ}(F) \simeq \mathbf{1}$.*

Proof. *(1)* By (A3) and Theorem 4.5, $\sigma^2 = \sigma + 1$ with $\sigma > 1$ gives $\sigma = \varphi$.

(2) By (A4) and Theorem 6.1, the admissible cost functional equals J_{cost} on $\mathbb{R}_{>0}$.

(3) By (A5) and Theorem 7.3, cost-minimum semantics and cost uniqueness force $r(s) = 1$ for all s, hence μ_F is uniform.

(4) By Theorem 2.7, uniform observables imply $\mathrm{SQ}(F)$ is a subsingleton. Since F is inhabited (A1), $\mathrm{SQ}(F)$ is nonempty. A nonempty subsingleton is equivalent to $\mathbf{1}$. □

Corollary 8.2 (Framework Equivalence)**.** *Let $\mathcal{C} = (\mathbf{1}, \mathrm{id}, \mathcal{O}_{\text{dim}}, \mu_{\mathcal{C}})$ be the canonical one-state framework. If F satisfies* (A1)–(A5) *and there exists an observable-type equivalence $e : O_F \simeq \mathcal{O}_{\text{dim}}$, then*

$$\mathrm{SQ}(F) \simeq \mathrm{SQ}(\mathcal{C}) \simeq \mathbf{1} \qquad \textit{and} \qquad O_F \simeq O_{\mathcal{C}}.$$

Remark 8.3 (What the theorem does not claim). The theorem does *not* assert that the raw state space S_F is a singleton. It asserts that all internal states are observationally indistinguishable at the cost-extracted interface: multiplicity of S_F is gauge redundancy, not physical freedom.

9 The Open Derivation Programme: Six Bridges

The full inevitability claim—"any $F \in \mathcal{Z}$ is forced into the RS structure"—requires deriving (A3)–(A5) from (A1)–(A2) alone. This section names the six structural bridges.

9.1 Bridge B1: self-similarity from zero parameters

Theorem 4.5 proves: *among* self-similar schemas, φ is forced. It does not prove: a zero-parameter ledger *must* be self-similar.

Open Problem 9.1 (Hierarchy Theorem)**.** Show that a discrete, zero-parameter ledger with at least two levels of event composition and uniform inter-level scaling necessarily satisfies a monic integer recurrence. This, combined with Theorem 4.5, forces φ.

The intermediate result "(A2) + hierarchical structure ⇒ (A3)" is well-posed. Deriving hierarchical structure from (A2) alone remains open.

9.2 Bridge B2: the RCL from ledger structure (deepest)

The RCL (1) is adopted as an axiom in (A4). Section 5 showed it is the d'Alembert equation—not a bespoke formula. The open problem is to *derive* it:

Open Problem 9.2 (Composition Rule Classification)**.** Among all continuous, symmetric composition rules $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ satisfying $J(xy) + J(x/y) = f(J(x), J(y))$ for a reciprocally symmetric cost J with $J(1) = 0$:

1. $f(0, 0) = 0$ and $f(a, 0) = 2a$ (from setting $y = 1$).
2. If cost composition is associative (derivable from the group structure of $(\mathbb{R}_{>0}, \times)$), then $f(a, b) = 2(a + 1)(b + 1) - 2$, i.e., J satisfies the RCL.

Prove or disprove that associativity is forced by the zero-parameter posture.

This is the most significant open problem in the programme.

9.3 Bridge B3: the ratio-extraction interface

(A5) posits $r : S_F \to \mathbb{R}_{>0}$. This bundles three sub-claims:

1. *Dimensionlessness:* all observables are ratios (no dimensionful constants in a zero-parameter framework).
2. *One-dimensionality:* a single real per state (requires the ledger to have one independent conserved quantity).
3. *Positivity:* the ratio lives on $\mathbb{R}_{>0}$.

9.4 Bridge B4: the ground-state restriction

The hypothesis $J(r(s)) = 0$ for all s is not automatic. By reciprocal symmetry, $r(S_F)$ is inversion-closed. Under a *structural determinacy* (SD) hypothesis—that each ratio is uniquely determined by a computable, $\mathbb{Z}/2\mathbb{Z}$-invariant predicate—any $x \neq 1$ requires at least 1 bit to select from $\{x, x^{-1}\}$, contradicting zero parameters. Deriving (SD) from (A2) alone is open.

Example 9.3 (Concrete falsifier for automatic ground-state)**.** Let $S_F = \{s_0\}$, $r(s_0) = 2$. Then $J_{\text{cost}}(r(s_0)) = J_{\text{cost}}(2) = \frac{1}{4} \neq 0$. This one-state framework satisfies (A1), (A2), (A4), and vacuously (A5-i), but violates the ground-state restriction.

9.5 Bridge B5: the constructive prediction map

Theorem 8.1 proves μ_F is constant but does not identify the constant. A prediction map $\mathcal{P} : (J_{\text{cost}}, \varphi) \to \mathcal{O}_{\text{dim}}$ would need to: (i) take J_{cost} and φ as sole inputs, (ii) produce specific dimensionless quantities, (iii) have $O(1)$ description length, and (iv) be unique. Existence (i)–(iii) can be established by formalizing the RS ledger algorithm; uniqueness (iv) and value identification remain open.

9.6 Bridge B6: the observable-type equivalence

Theorem 8.2 requires $e : O_F \simeq \mathcal{O}_{\text{dim}}$ as input. After quotient collapse, $\text{im}(\mu_F)$ is a singleton. Any two singleton types are abstractly isomorphic, so the type-level equivalence is trivially constructible. Non-trivial content (value identification) reduces to B5.

10 Falsifiers

The theorem is a mathematical statement that cannot be refuted within its hypotheses. However, the *inevitability framing* can be challenged by showing that a hypothesis is too strong, too narrow, or physically unwarranted. Each of the following would constitute a substantive refutation:

1. **Alternative zero-parameter framework.** Construct $F \in \mathcal{Z}$ that derives observables and does *not* collapse on the quotient (non-uniform μ_F without free parameters).
2. **Non-φ self-similarity.** Provide a parameter-free self-similar scaling whose preferred scale is not φ, using a notion of self-similarity that retains the intended physical meaning.
3. **Continuous zero-parameter physics.** Exhibit a theory with no adjustable dimensionless knobs whose state space is genuinely uncountable, and argue that continuity does not constitute hidden parameter freedom.

4. **Conservation without double-entry.** Produce a discrete conserved dynamics where conservation is representable without a balanced accounting structure.

5. **Pathological d'Alembert solution.** Construct a non-standard solution to (3) satisfying the raw identity but violating regularity. This would not refute Theorem 6.1 (regularity is explicit), but would show that regularity is a load-bearing hypothesis.

11 Conclusion

We have proved that any physics framework satisfying the five admissibility conditions has a unique observational state on the quotient: $\mathrm{SQ}(F) \simeq \mathbf{1}$. The preferred scale is $\varphi = (1+\sqrt{5})/2$ and the cost functional is $J_{\mathrm{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Three of the five assumptions are adopted axioms, not yet derived from the zero-parameter posture alone. The paper names six bridges (B1–B6) that would close the full inevitability programme. Of these, B2 (deriving the RCL/d'Alembert equation from ledger structure) and B5 (constructing a unique prediction map) are the deepest open problems.

The composition law is the d'Alembert functional equation (Section 5)—the unique symmetric composition law on $(\mathbb{R}_{>0}, \times)$ compatible with non-degeneracy. It is not bespoke. Whether it is *forced* by the zero-parameter posture is the central question the programme must answer.

A Formal Verification

All proved results are formally verified in Lean 4. The following table maps paper results to Lean modules. The Lean codebase uses one external axiom: `aczel_representation_axiom` (the Aczél classification, Theorem 5.3).

Paper result	**Lean theorem**	**Module**
Theorem 4.5 (φ forcing)	`phi_forced`	`PhiForcing`
Theorem 6.1 (cost uniqueness)	`T5_uniqueness_complete`	`CostUniqueness`
Theorem 7.3 (derived uniformity)	`zero_params_forces_uniform`	`ModelIndependent`
Theorem 8.1 (quotient collapse)	`derived_model_independent_exclusivity`	`ModelIndependent`
Theorem 8.2 (framework equivalence)	`model_independent_framework_equiv`	`ModelIndependent`
B1 hierarchy theorem	`hierarchy_forces_fibonacci_recurrence`	`HierarchyTheorem`
B4 ground-state corollary	`ground_state_implies_cost_minimum`	`GroundStateRestriction`
B6 type equivalence	`bridge_B6_observable_type_equiv_trivial`	`ObservableTypeEquiv`

All modules listed above compile with zero `sorry`. The B2 scaffold (`RCLDerivation`) and B5 scaffold (`PredictionMap`) contain explicit `sorry`-marked open problems corresponding to Theorem 9.2 and Section 9.5.

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